C 23897 **(Pages : 2)** **Name.....**

Reg. No.....

SECOND SEMESTER (CBCSS–UG) DEGREE EXAMINATION, APRIL 2022

Statistics

STA 2C 02—PROBABILITY THEORY

(2019—2020 Admissions)

Time : Two Hours Maximum : 60 Marks

Use of Calculator and Statistical tables are permitted.

Part A (Short Answer Type Questions)

Each question carries 2 *marks.*

Maximum marks that can be scored from this part is 20.

1. Show that the probability of an impossible event is zero.
2. If A,B and C are 3 mutually independent events, then show that $A \cup B$ and C are also independent.
3. State multiplication theorem of probability.
4. Explain the terms : (i) Sample space ; (ii) Mutually exclusive events ; (iii) Exhaustive events ; and (iv) Equally likely events.
5. Define random variable. Distinguish between discrete and continuous random variable using examples.
6. Let X be a random variable with p.d.f. $f(x) = (1/2)^x$, $x = 1,2,...$ and zero elsewhere. Find the p.d.f of $Y = X^2$.
7. For the density function $f(x) = ke^{-\theta x}$, $x \geq 0, \theta > 0$ and 0 elsewhere, find the value of k.
8. If the first three moments about origin are respectively 1,7 and 38. Obtain the co-efficient of skewness.
9. If $\mu'_r = r!$ for a random variable X, find its m.g.f.
10. For two dimensional discrete random variables X and Y show that E (X + Y) = E (X) + E (Y).
11. For two random variables X and Y show that V (Y) = E [V(Y/X)] + V [E(Y/X)].
12. Define conditional mean and conditional variance.

Turn over

Part B (Short Essay/Paragraph Type Questions)

Each question carries 5 marks.
Maximum marks that can be scored from this part is 30.

13. Two fair dice are thrown once. Find the probability that sum of the upper faces is 10 or above provided first die shows an even number.

14. Two urns I and II contain respectively 3 white and 2 black balls, 2 white and 4 black balls. One ball is transferred from urn 1 to the urn II and then a ball is drawn from the latter. It happens to be white. What is the probability that the transferred ball was white ?

15. What are the properties of p.d.f and distribution function?

16. A coin is known to come up heads three times as often as tails. This coin is tossed three times. Let X be the number of heads that appear. Write down the possibility density function of X.

17. Define skewness and kurtosis. Give any one measure for examining skewness and kurtosis.

18. State and prove Cauchy-Schwartz inequality for two random variables X and Y.

19. For the joint function, $f(x, y) = \frac{2}{3}(1+x)e^{-y}, 0 < x < 1, y > 0.$ Obtain the conditional distribution of x given $y = 1$ and that of y given $x = 1/3$.

Part C (Essay Type Questions)

Answer any **one** *question.*
The question carries 10 *marks.*
Maximum marks that can be scored from this part is 10.

20. (i) State and prove addition theorem of probability.

    (ii) The probability of male birth is 0.6.Find the probability that in four births : (a) All are boys ; (b) All are girls ; (c) Atleast one is a girl ; and (d) There are two boys and two girls.

21. Two random variables X ang Y have the following p.d.f :

    $f(x, y) = 2 - x - y, 0 \le x \le 1, 0 \le y \le 1$ and zero elsewhere. Find :

    (i) Marginal probability density functions of X and Y.

    (ii) Conditional density functions.

    (iii) Var(x) and Var(Y).